

Ex 21.

$$H(z) = \frac{1}{(z-0.2)(z-0.9)} \quad (T=1)$$

$$\text{men } H(e^{j\omega T}) = \frac{1}{(e^{j\omega T}-0.2)(e^{j\omega T}-0.9)}$$

$$|H(e^{j\omega T})| = 4.10$$

$$\angle H(e^{j\omega T}) = -1.7586 \quad (\text{rad})$$

$$y|_{k=20} = 5 \times 4.10 \cdot \cos(0.3k + 0.1 - 1.7586)$$

$$= 20.50 \cdot \cos(0.3k - 1.6586) \quad \text{för } k \text{ stor}.$$

Ex 22.

$$\begin{aligned} G_{ZAS}(z) &= \mathcal{Z} \left[\mathcal{L}^{-1} \left(\frac{1-e^{Ts}}{s} G(s) \right) \right] \quad \text{ZOH} \\ &= (1-z^{-1}) \cdot \mathcal{Z} \left\{ \frac{G(s)}{s} \right\} \\ &= (1-z^{-1}) \cdot \mathcal{Z} \left(\frac{1}{s^3} \right) \\ &= (1-z^{-1}) \cdot \frac{1}{2!} \cdot \frac{\partial^2}{\partial \alpha^2} \left(\frac{z}{z-e^{\alpha T}} \right) \\ &= \frac{z-1}{z} \cdot \frac{1}{2} \cdot \frac{\partial}{\partial \alpha} \left(\frac{z T e^{\alpha T}}{(z-e^{\alpha T})^2} \right) \\ &= \frac{1}{2} \cdot \frac{z-1}{z} \cdot \frac{T (T e^{\alpha T} (z-e^{\alpha T})^2 - e^{\alpha T} \cdot 2(z-e^{\alpha T}) \cdot (-T e^{\alpha T}))}{(z-e^{\alpha T})^4} \\ &= \frac{T}{2} \cdot (z-1) \cdot \frac{T(z-1)^2 + 2T(z-1)}{(z-1)^4} \\ &= \frac{T^2}{2} \frac{z+1}{(z-1)^2} \end{aligned}$$

$$\begin{aligned} \text{Ex 23. } H_{GAS}(z) &= \mathcal{Z} \left\{ \mathcal{L}^{-1} \left(\frac{1-e^{Ts}}{s} \cdot H(s) \right) \right\} \\ &= (1-z^{-1}) \mathcal{Z} \left(\frac{a}{s(s+a)} \right) \end{aligned}$$

$$\frac{A}{s} + \frac{B}{s+a}$$

$$= \frac{A(s+a) + Bs}{s(s+a)} = \frac{(A+B)s + Aa}{s(s+a)}$$

$$\Rightarrow \begin{cases} A+B=0 \\ A=1 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \end{cases}$$

$$\Rightarrow H_{GAS}(z) = (1 - z^{-1}) \cdot Z\left(\frac{1}{s} - \frac{1}{s+a}\right)$$

$$= \frac{z-1}{z} \cdot \left(\frac{z}{z-1} - \frac{z}{z-e^{-aT}}\right)$$

$$= 1 - \frac{z-1}{z-e^{-aT}}$$

$$= \frac{1-e^{-aT}}{z-e^{-aT}}$$

Ex 25. $G(s) = \frac{1}{s^2} = \frac{Y(s)}{U(s)} \Rightarrow s^2 Y(s) = U(s) \Rightarrow \ddot{y}(t) = u(t)$

Let $x_1 = y$, $x_2 = \dot{x}_1$, then

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u \end{cases} \Rightarrow \underline{\dot{X}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \underline{\dot{X}} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\underline{F}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\underline{X}} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\underline{G}} u$$

$$\underline{Y} = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\underline{H}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\underline{\Phi} = e^{\underline{E}T} = \underline{I} + \underline{E}T + \frac{\underline{E}^2 T^2}{2!} + \dots$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & T \\ 0 & 0 \end{bmatrix} + 0 + \dots = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}$$

$$\underline{\Gamma} = \int_0^t e^{\underline{E}\tau} d\tau \underline{G} = \int_0^t \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} d\tau \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \tau & \frac{1}{z}\tau^2 \\ 0 & \tau \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{z}\tau^2 \\ \tau \end{bmatrix}$$

$$G_{us}(z) = H[zI - \Phi]^{-1} \Gamma$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \left[\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \right]^{-1} \begin{bmatrix} \frac{1}{z}\tau^2 \\ \tau \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} z-1 & -\tau \\ 0 & z-1 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{z}\tau^2 \\ \tau \end{bmatrix}$$

$$= \frac{\tau^2}{z} \frac{z+1}{(z-1)^2}$$

Ex 26. BIBO stability

$$u_k = a u_{k-1} + b e_k$$

Input : impulse : $e_k = \delta_k$, i.e., $e_0 = 1$, others = 0

Init. condi. : $u_{-1} = u_{-2} = \dots = 0$ (zero init)

then:

$$u_0 = 0 + b = b$$

$$u_1 = a u_0 + 0 = ab$$

$$u_2 = a u_1 + 0 = a^2 b$$

$\vdots$

$$u_k = a^k b$$

$$V(z) = H(z) \cdot \underbrace{E(z)}_1 = H(z) \Rightarrow u_k = h_k$$

For BIBO stability: $\sum_k |h_k| < +\infty$

then: $\sum_{k=0}^{\infty} |h_k| = \sum_{k=0}^{\infty} |a^k b| \leq |b| \sum_{k=0}^{\infty} |a^k| = |b| \cdot \frac{1-|a|^{\infty}}{1-|a|} < \infty$

$\Rightarrow |a| < 1$

Ex 27.

trapezoid.

$$u_k = u_{k-1} + \frac{T}{2} (e_k + e_{k-1})$$

calculate response to $e = \delta$, then:

$$u_0 = u_{-1} + \frac{T}{2} (1 + 0) = \frac{T}{2}$$

$$u_1 = u_0 + \frac{T}{2} (0 + 1) = T$$

$$u_2 = u_1 + \frac{T}{2} (0 + 0) = T$$

$\vdots$

$$u_k = T$$

$$\sum_{k=0}^{\infty} |h_k| = \sum_{k=0}^{\infty} |u_k| = \frac{T}{2} + \sum_{k=1}^{\infty} T = +\infty \quad \text{NOT BIBO stable.}$$

its transfer function:

$$V(z) = z^{-1} V(z) + \frac{T}{2} (E(z) + z^{-1} E(z))$$

$$\Rightarrow H(z) = \frac{V(z)}{E(z)} = \frac{T}{2} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{T}{2} \frac{z - 1}{z + 1} \Rightarrow \underline{p = -1}$$

(not)

Ex 28.

$$z = p - N = 0 - 0 = 0 \quad \text{stable}$$

Ex 34.

$$G(s) = \frac{K_m}{s(s+1)}$$

$$G_{ZAS}(s) = (1 - z^{-1}) z^{-1} \left(\frac{K_m}{s(s+1)} \right)$$

$$\frac{10}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

$$z^{-1} \left(\frac{1}{s(s+1)} \right) = \frac{z}{z-1} - \frac{z}{z - e^{-10T}}$$

$$G_{ZAS}(s) = 1 - \frac{z-1}{z - e^{-10T}} = \frac{1 - e^{-10T}}{z - e^{-10T}} = \frac{0.289}{z - 0.711}$$

Try PI controller,

$$C(z) = K_p + \frac{K_i z}{z-1}$$

$$CG = \frac{(K_p + K_i)z - K_p}{z-1} \cdot \frac{0.289}{z - 0.711}$$

$$= \frac{0.289(K_p + K_i)z - 0.289K_p}{z^2 - 1.711z + 0.711}$$

$$\rho(z) = \frac{CG}{1 + CG} = \frac{0.289(K_p + K_i)z - 0.289K_p}{z^2 - 1.711z + 0.711 + 0.289(K_p + K_i)z - 0.289K_p}$$

$$= \frac{0.289(K_p + K_i)z - 0.289K_p}{z^2 + [0.289(K_p + K_i) - 1.711]z + 0.711 - 0.289K_p}$$

$$\Rightarrow \begin{cases} 0.289(K_p + K_i) - 1.711 = -2\alpha(\omega_d T) e^{-\xi\omega_n T} = -0.90334 \\ 0.711 - 0.289K_p = e^{-2\xi\omega_n T} = 0.303 \end{cases}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 17.854 \text{ rad/s}$$

$$\Rightarrow \begin{cases} K_P = 1.412 \\ K_I = 1.383 \end{cases}$$

$$\text{Therefore, } C(z) = \frac{(K_P + K_I)z - K_P}{z-1} = \frac{2.79z - 1.41}{z-1}$$

$\eta = 1$, no ess for unit step input.

Ex 36.

$$G_{ZAS} = (1 - z^{-1}) z \left(\frac{1}{s(s+1)(s+3)} \right)$$

$$\begin{cases} A = \frac{1}{3} \\ B = -\frac{1}{2} \\ C = \frac{1}{6} \end{cases}$$

$$z(\) = z \left(\frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3} \right)$$

$$= \frac{1}{3} \cdot \frac{z}{z-1} - \frac{1}{2} \frac{z}{z-e^{-T}} + \frac{1}{6} \frac{z}{z-e^{-3T}}$$

$$G_{ZAS} = \frac{1}{3} - \frac{1}{2} \frac{z-1}{(z-e^{-T})} + \frac{1}{6} \frac{z-1}{(z-e^{-3T})}$$

$$= \frac{2 - 3(z-1)(z-e^{-3T}) + (z-1)(z-e^{-T})}{6(z-e^{-T})(z-e^{-3T})}$$

$$= \frac{(-3+1)z^2 - 3(-1-e^{-3T})z + (-1-e^{-T})z + 2 + e^{-3T} + e^{-T}}{6(z-e^{-T})(z-e^{-3T})}$$

$$= \frac{-2z^2 + 3.808z + 3.903}{6(z-0.97)(z-0.92)}$$

$$= 0.3 \frac{z^2 - 1.904z + 1.9515}{(z-0.97)(z-0.92)}$$

$$P_0 = 100 \cdot e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} = 5$$

$$\Rightarrow \zeta = 0.640$$

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Ex 37.

$$\dot{y}_k \approx \frac{y_k - y_{k-1}}{T}$$

$$\Rightarrow sY(s) \leftrightarrow \frac{Y(z) - Y(z) \cdot z^{-1}}{T}$$

$$\Rightarrow sY(s) \leftrightarrow \frac{z-1}{Tz} \cdot Y(z) \quad \text{backward}$$

$$\Rightarrow s \leftrightarrow \frac{z-1}{Tz}$$

$$y_k = y_{k-1} + \frac{T}{2} (u_k + u_{k-1})$$

$$\Rightarrow Y(s) = \frac{U(s)}{s} \Leftrightarrow Y(z) = Y(z) \cdot z^{-1} + \frac{T}{2} (U(z) + U(z) \cdot z^{-1})$$

$$Y(s) = \frac{U(s)}{s} \Leftrightarrow Y(z) = \frac{T}{2} \frac{1+z^{-1}}{1-z^{-1}}$$

$$\Rightarrow s \leftrightarrow \frac{2}{T} \frac{z-1}{z+1} \quad \text{Trapezoid Rule.}$$

Ex 38.

$$H(s) = \frac{U(s)}{E(s)} = \frac{a}{s+a} \Rightarrow \dot{u} + au = ae$$

$$\Rightarrow \dot{u} = a(e - u)$$

$$u = \int_0^t a(e - u) dt$$

$$\Leftrightarrow u_k - u_{k-1} = a(e_{k-1} - u_{k-1}) \cdot T$$

$$\Leftrightarrow U(z) - U(z) \cdot z^{-1} = a(E(z) - U(z)) \cdot z^{-1} \cdot T$$

$$\Leftrightarrow \frac{U(z)}{E(z)} = \frac{aT}{z^{-1} + aT} = \frac{a}{\frac{z-1}{T} + a}$$

